ACKNOWLEDGEMENT

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EXAMINER 1 **EXAMINER 2**

1.INTRODUCTION

I am delighted to present my formal internship report, documenting my immersive and rewarding experience as an intern at **Eagle-Hitech Softclou Pvt. Ltd.**

This report aims to provide a comprehensive overview of the projects I worked on, the skills I acquired, and the valuable insights I gained during my internship period.

I pursued this internship at **Eagle-Hitech Softclou Pvt. Ltd**. Because of its renowned reputation and consistent pursuit of quality and innovation.

Securing a position here was a competitive process, and I am immensely grateful for the opportunity to be part of such a talented team.

At the onset of the internship, my primary expectations were to gain hands-on experience working on real-world projects, refine my technical skills, and immerse myself in a professional environment.

I also set personal goals to learn from experienced mentors, collaborate effectively with the team, and contribute meaningfully to the success of the projects I was involved in.

2.COMPANY PROFILE

Eagle-Hitech Softclou Pvt Ltd is one of the leading bootstrapped pure-play IT Services and Software development company that are on a mission to provide an industry-leading platform for software applications.

Eagle-Hitech Softclou Pvt Ltd is constantly expanding its family and is looking for dynamic fresher's and graduates with a background in Computer Science.

Once a part of the family, you can be confident that your career is in safe hands. **Eagle-Hitech Softclou Pvt Ltd** has clients all over the world that hands you the opportunity to work on a variety of projects.

If you believe testing to be your forte, then **Eagle-Hitech Softclou Pvt Ltd** is one of the best places you could possibly work in.

With more than 12 Years of industry expertise, **Eagle-Hitech Softclou Pvt Ltd** has become the undisputed market leader of lending IT Services & automation systems.

We relentlessly strive in providing secure and scalable lending solutions to a broader financial spectrum. Today, we are a company with 80+ professionals and 100+ happy clients. We have operations in India.



2.1 HISTORY ABOUT THE COMPANY

Eagle-HiTech Softclou Private Limited in Kodambakkam, Chennai has earned a stellar reputation for providing top-notch training services.

Founded in 07-2012, the company boasts a 4.9 rating based on 1627 reviews, making it a trusted name in the industry.

Situated near Opposite to Agarwal Hospital, the facility is easily accessible to individuals in Chennai and nearby areas.

The company is known for its state-of-the-art infrastructure, equipped with modern training facilities. The learning environment at **Eagle-HiTech Private Softclou Limited** in Kodambakkam, Chennai is designed to foster growth and creativity, providing participants with the ideal atmosphere to focus on their professional development.

Eagle-HiTech Softclou Private Limited, located in Kodambakkam, Chennai, has been a leader in skill development for many years. The company specializes in providing a comprehensive range of training programs designed to equip individuals with the practical knowledge and expertise needed to excel in the chosen fields.



PROGRAMS OFFERED :

Eagle-HiTech Softclou Private Limited in Kodambakkam, Chennai offers a variety of skill development programs to meet the evolving demands of various industries.

They offer many high demanding courses such as :

- Cyber Security
- Full Stack
- Full Stack Development
- Hardware Engineering
- Hardware Networking
- C Programming
- Machine Learning With Python
- Oracle Developer
- UI UX Design
- Cloud Computing



SERVICES:

We offer a comprehensive suite of IT services designed to meet the diverse needs of businesses.

IT Solutions Tailored for Your Success:

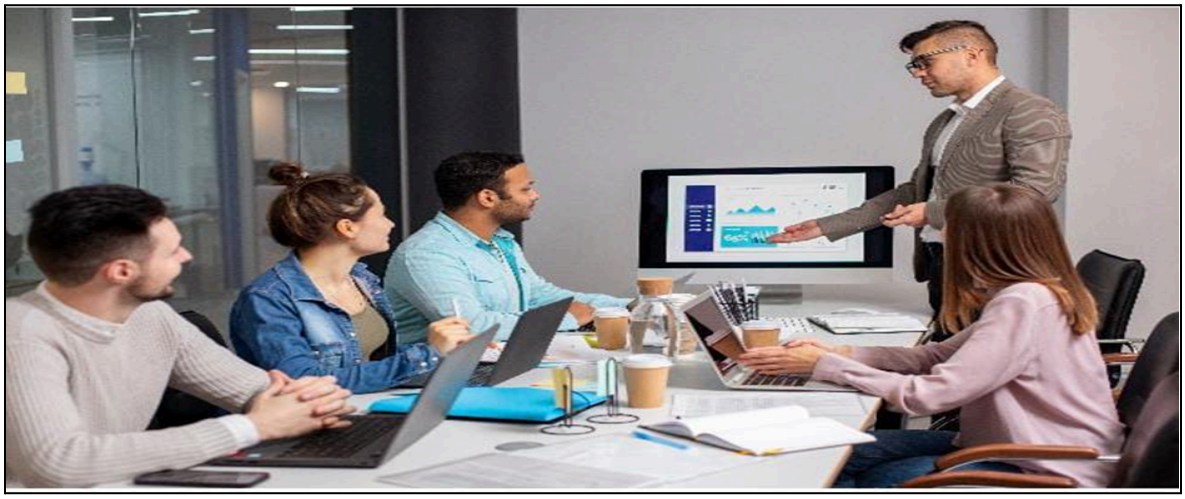
1.Consulting and Outsourcing:

Our consulting and outsourcing services provide expert guidance and efficient solutions to streamline your operations and drive business growth.



2.Development and Maintenance:

Development and Maintenance services ensure seamless enhancement and reliable upkeep of your IT systems and automation solutions, driving continuous performance and adaptability to your evolving business needs.



3. Technology Expertise:

Leverage our deep technology expertise to deliver innovative IT solutions and automation systems tailored to optimize your business operations.



4. Research and Development:

Our Research and Development service drives innovation by exploring and implementing cutting-edge IT solutions and automation technologies to keep your business ahead of the curve.



PROCESS AND BENEFITS:

1.GATHERING REQUIREMENTS

In our first meetings, we focus on understanding your vision. Based on this, we develop a strategy, select the most suitable technology, set timelines, and assign team members to the project.

2.UX/UI

We begin by creating initial designs that encapsulate the visual identity of your app. By prioritizing the needs of your users, we develop attractive and user-friendly interfaces that enhance your brand, foster customer loyalty, and facilitate business expansion.

3.DEVELOPMENT

In practical terms, theory is put into action by creating the solution along with its associated systems, security measures, user permissions, connections, and functionalities.

4.TESTING

A series of tests are conducted to guarantee the stability and security of your solution. We typically opt for actual devices over emulators when collecting and evaluating feedback from test users.

5.SUPPORT AND MAINTENANCE

There is no end to innovation. Even after we provide the solution, we are available for updates and regular maintenance.

3. WORK PROCESS

During my internship at ETS Softclou Pvt. Ltd., I worked on a Data-Driven Project focused on **Detection of Deceptive Job Postings Using Machine Learning**. The goal was to develop a Machine Learning Classification Model using natural language processing (NLP) techniques. I applied TF-IDF vectorization to job descriptions and trained a logistic regression model to classify postings as either genuine or fake.

3.1 TOOLS USED : GOOGLE COLLAB

- Google Colab is a free, cloud-based platform provided by Google that allows you to write and run Python code in a web browser. It is especially useful for machine learning, data science, and deep learning projects, and provides free access to GPUs and TPUs.
- Colab runs entirely in your web browser, eliminating the need to download and install Python or any other software on your computer.
- Colab is built on top of Jupyter Notebooks, which are interactive documents that combine code, text, and visualizations.
- Colab notebooks can be easily shared with others, making it convenient for collaboration on coding projects.
- Colab integrates seamlessly with Google Drive, allowing you to store and manage your notebooks in your Google Drive account.
- Colab is widely used for learning Python, exploring data science concepts, and building machine learning models.



3.2 IMPORTING LIBRARIES :

Deceptive Job Postings Dataset(Classification):

The project involves analysis of fake jobs in the dataset .The main steps include data cleaning, visualization, feature extraction, model training, and evaluation using confusion matrix and classification metrics.

- ☐ Pandas
- ☐ Matplotlib
- ☐ Seaborn
- ☐ Scikit-learn

PANDAS :

- Pandas is an open-source Python library used for data manipulation and analysis.
- It provides powerful tools like **DataFrame** and **Series** for handling structured data.
- It supports operations such as filtering, grouping, merging, and reshaping data.
- Pandas is widely used in data science, machine learning, and data preprocessing tasks.

MATPLOTIB :

- Matplotlib is an open-source Python library for creating static, animated, and interactive visualizations.
- It supports various plot types like line, bar, scatter, and pie charts.
- It allows customization of colors, labels, and styles for better presentation.
- Matplotlib is widely used in data analysis to visually represent trends and patterns.

SEABORN :

- Seaborn is an open-source Python library built on top of Matplotlib for statistical data visualization.
- It provides a high-level interface for creating attractive and informative graphs.

- It includes built-in themes, color palettes, and functions for complex visualizations like heatmaps and violin plots.
- Seaborn is widely used in data analysis to explore and understand data patterns more easily.

SCIKIT-LEARN :

- Scikit-learn is an open-source Python library for machine learning and data analysis.
- It provides simple and efficient tools for tasks like classification, regression, clustering, and dimensionality reduction.
- It includes built-in datasets, model selection utilities, and evaluation metrics.
- Scikit-learn is widely used for building and testing machine learning models in both research and industry.



STEP 1 : LOAD DATASETS

To use an XLSX file in Python, you can load it with Pandas using the **read_excel()** function.

```
df = pd.read_excel("fake_job_postings.xlsx")
```

3.3 BASIC DATA INSPECTION :

head():

- Displays the first few rows of the DataFrame (default is 5).
- Useful for quickly previewing how the data looks.
- Helps check column names, data types, and value formatting.

tail():

- Displays the last few rows of the DataFrame (default is 5).
- Useful for checking the end of the dataset for missing or odd values.
- Often used to spot formatting issues that occur at the bottom.

info():

- Prints a summary of the DataFrame, including column names and data types.
- Shows the number of non-null values in each column.

shape():

- Returns a tuple (rows, columns) representing dataset dimensions.
- Quickly tells you how big the dataset is.
- Often used before and after cleaning to see changes in size.

dtypes():

- Lists the data type of each column (object, int64, float64, etc.).
- Helps identify categorical vs. numerical vs. text columns.
- Useful for deciding preprocessing steps.

dropna():

- Removes rows (or columns) with missing values (NaN).
- Prevents errors during analysis and model training.

duplicated() & drop_duplicates():

- duplicated(): Checks for rows that are exact copies of previous rows.
- drop_duplicates(): Removes duplicate rows from the DataFrame.

3.4 MODELLING AND EVALUATION METHODOLOGY :

This internship project involved a systematic approach combining data preprocessing, feature extraction, model training, and performance evaluation. The key methodologies applied are outlined below:

1. TF–IDF (Term Frequency – Inverse Document Frequency)

TF–IDF is a statistical method used to transform unstructured text into meaningful numerical features. It assigns a weight to each word based on its frequency within a document relative to its frequency across the entire corpus. This approach reduces the impact of common words while giving greater importance to rare yet informative terms.

- Logarithmic Scaling: Applied in the Inverse Document Frequency (IDF) calculation to dampen the influence of highly frequent words.
- Frequency Ratios: Measure how often a term appears in a document compared to its occurrence in the corpus, helping to distinguish informative from generic words.
- Sparse Representation: The resulting TF–IDF matrix is high-dimensional and mostly composed of zeros, making it efficiently stored and processed in sparse form.

2. Train–Test Split

The dataset was divided into training and testing subsets to evaluate the model's ability to generalize to unseen data. This ensures that the model's performance is not solely based on memorizing training examples.

- Probability-Based Sampling: Data points were split randomly, giving each record an equal chance of selection.
- Reproducibility: A fixed random seed was used to maintain consistent results across multiple runs.

3. Logistic Regression

Logistic Regression, a widely used algorithm for binary classification, was employed to distinguish between real and fraudulent job postings. It models the relationship between input features and the probability of a class using a logistic (sigmoid) function.

- Linear Algebra Foundation: Input features are combined linearly using learned weights and a bias term to define the decision boundary.
- Sigmoid Function: Transforms linear combinations into probabilities ranging from 0 to 1.
- Optimization & Maximum Likelihood Estimation (MLE): Model parameters are updated iteratively using optimization techniques to maximize the likelihood of correctly predicting the observed data.

4. Classification Metrics

Model performance was evaluated using metrics such as precision, recall, F1-score, and accuracy, which provide a comprehensive understanding of classification quality.

- Conditional Probability Basis: Precision and recall are derived from probabilities that quantify the likelihood of correct predictions.
- Ratio Metrics: These metrics express performance as proportions, offering insights into trade-offs between identifying true positives and avoiding false positives.

5. Confusion Matrix

The confusion matrix is a tabular summary that compares predicted labels to actual labels, offering a clear view of classification outcomes.

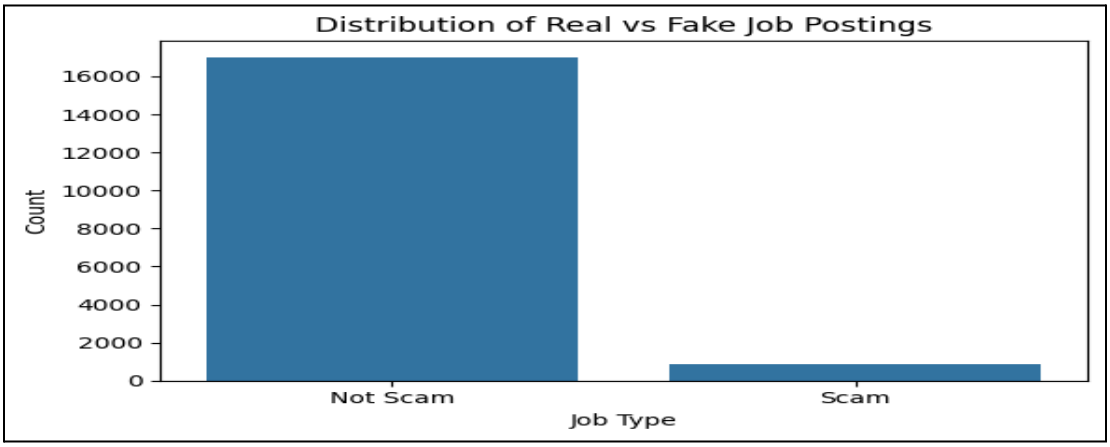
- Counting Mechanism: Records true positives, false positives, true negatives, and false negatives directly from predictions.
- Visual Interpretation: The matrix is represented both textually and as a heatmap, making it easier to identify error patterns and model strengths.

3.5 EXPLORATORY DATA ANALYSIS :

Exploratory Data Analysis (EDA) was performed to better understand the structure and patterns within the job postings dataset.The focus was on analyzing the textual descriptions and how they differ between real and fraudulent job postings.By exploring features like description length and word count, we aimed to uncover patterns that may indicate fraud.These insights guided feature selection and supported the model-building process.

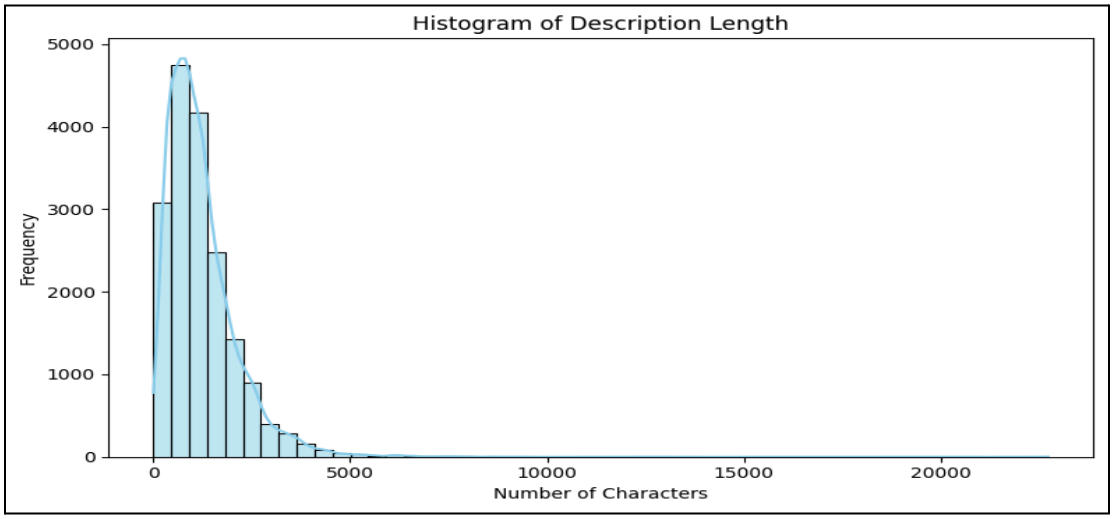
1.Class Distribution Plot

- Helps visualize the balance or imbalance between real and fake job postings.
- Useful for detecting class imbalance, which can affect model performance and bias.



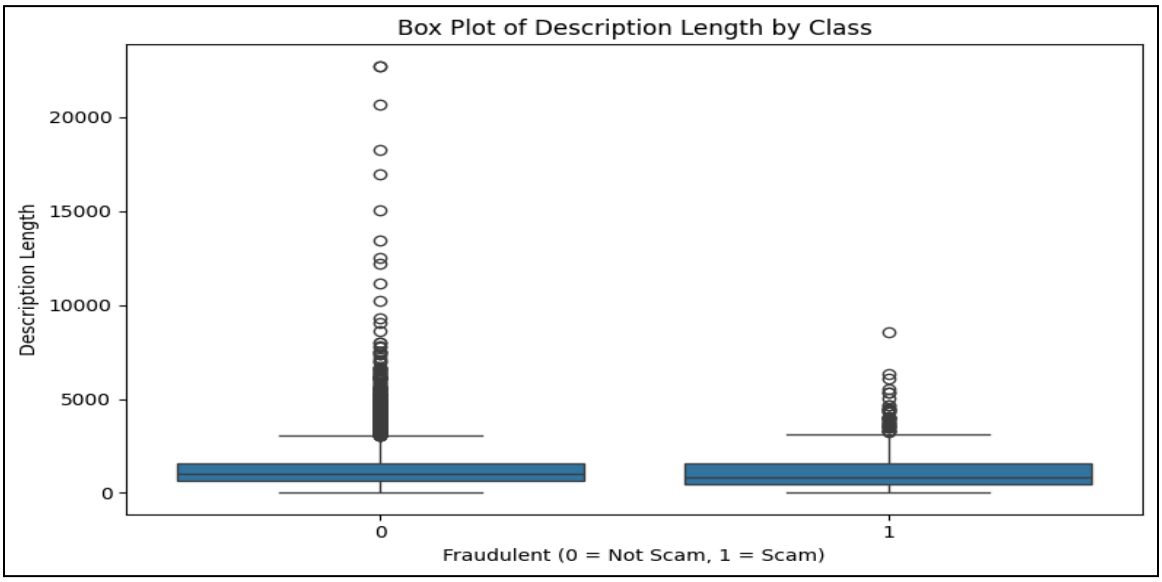
2.Histogram of Description Length

- Shows the frequency distribution of text length (in characters) across job descriptions.
- Helps identify outliers or common lengths, e.g., unusually short or long descriptions.



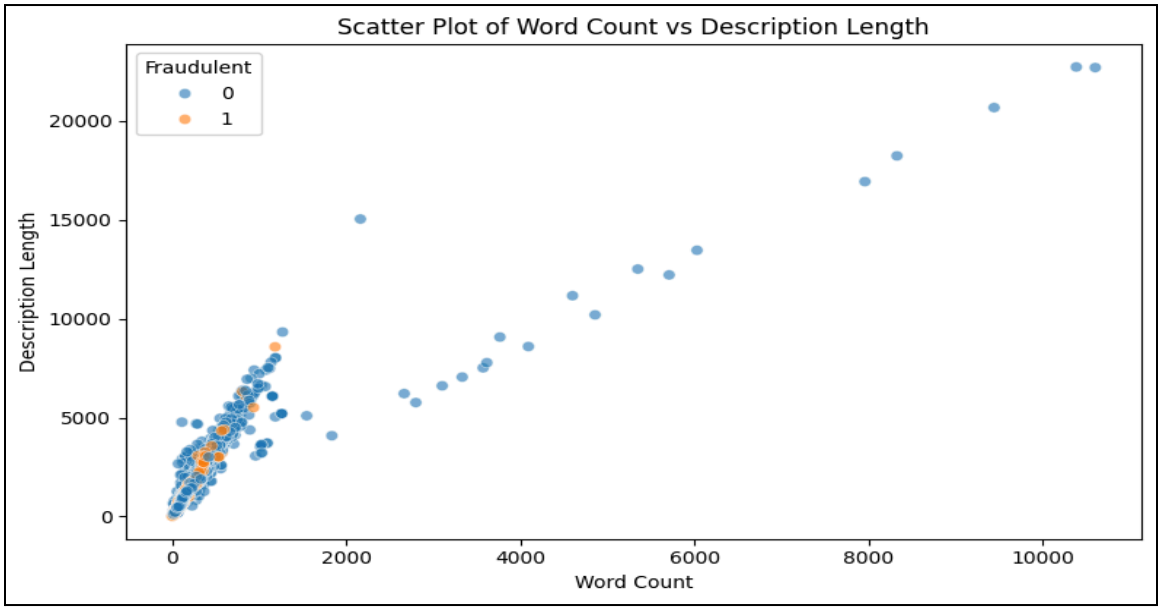
3.Box Plot of Description Length by Class

- Compares the spread and central tendency of description lengths for scam vs non-scam jobs.
- Highlights outliers and potential differences in writing patterns between classes.
- Highlights outliers and potential differences in writing



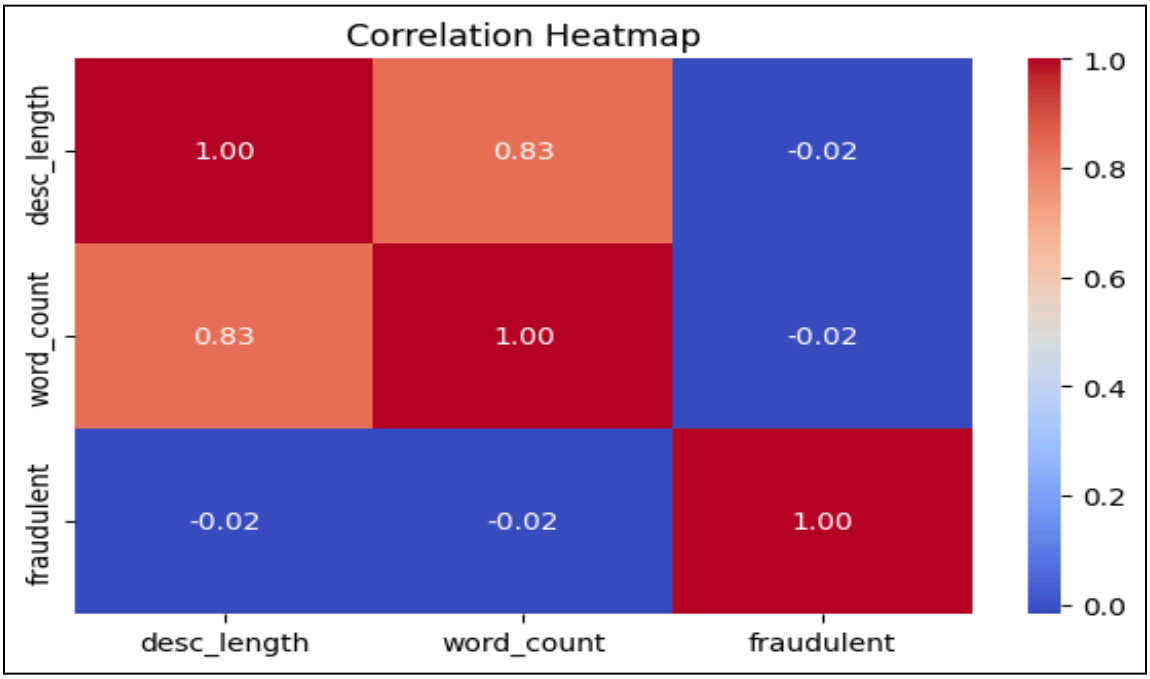
4.Scatter Plot (Word Count vs Description Length)

- Displays the relationship between number of words and characters in descriptions.
- Colored by class, helping to see if scam jobs cluster differently from real jobs.



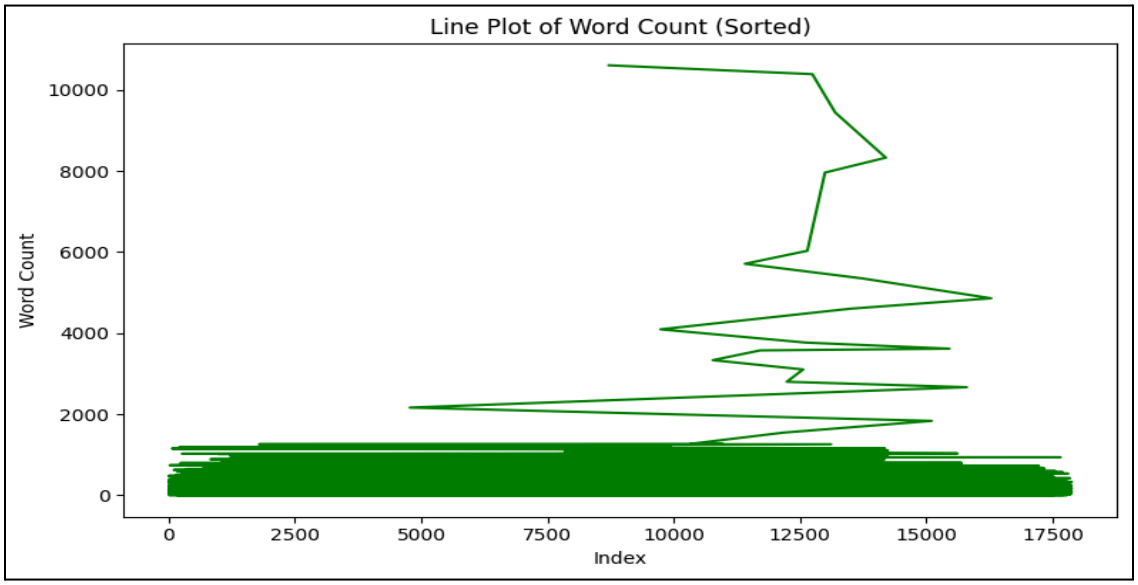
5.Correlation Heatmap

- Shows correlation coefficients between numerical features (desc_length, word_count, fraudulent).
- Helps detect linear relationships that might influence feature importance in modeling.



6.Line Plot of Word Count (Sorted)

- Visualizes the trend of word counts across all sorted job descriptions.
- Good for spotting gradual changes or spikes, and understanding data density.



REPORT

4.1 DATASET INSPECTION REPORT :

Basic information about the Dataset.

```
========================================
BASIC DATA INSPECTION
========================================

Dataset Shape: 17880 rows, 18 columns

Data Types:
job_id                  int64
title                  object
location               object
department             object
salary_range           object
company_profile        object
description            object
requirements           object
benefits               object
telecommuting           int64
has_company_logo        int64
has_questions           int64
employment_type        object
required_experience    object
required_education     object
industry               object
function               object
fraudulent              int64
```

HANDLING MISSING VALUES AND DUPLICATES :

```
# Step 1.1: Handle missing values and duplicates
print(f"Duplicate rows before dropping: {df.duplicated().sum()}")
df = df.drop_duplicates()
print(f"Dataset shape after dropping duplicates: {df.shape}")
```

```
# Step 2: Handling missing values :
df = df[['description', 'fraudulent']].dropna()
```

```
Duplicate rows before dropping: 0
Dataset shape after dropping duplicates: (17880, 18)
```

DATASET INFO :

```
========================================
BASIC DATA INSPECTION
========================================

Dataset Shape: 17880 rows, 18 columns

Data Types:
job_id                  int64
title                  object
location               object
department             object
salary_range           object
company_profile        object
description            object
requirements           object
benefits               object
telecommuting           int64
has_company_logo        int64
has_questions           int64
employment_type        object
required_experience    object
required_education     object
industry               object
function               object
fraudulent              int64
```

FIRST 3 ROWS:

```
--- First 3 Rows ---
   job_id                                  title          location  \
0       1                          Marketing Intern  US, NY, New York
1       2  Customer Service - Cloud Video Production    NZ, , Auckland
2       3    Commissioning Machinery Assistant (CMA)     US, IA, Wever

  department salary_range                                company_profile  \
0  Marketing          NaN  We're Food52, and we've created a groundbreaki...
1    Success          NaN  90 Seconds, the worlds Cloud Video Production ...
2        NaN          NaN  Valor Services provides Workforce Solutions th...

                                        description  \
0  Food52, a fast-growing, James Beard Award-winn...
1  Organised - Focused - Vibrant - Awesome!Do you...
2  Our client, located in Houston, is actively se...

                                        requirements  \
0  Experience with content management systems a m...
1  What we expect from you:Your key responsibilit...
2  Implement pre-commissioning and commissioning ...

                                        benefits  telecommuting  \
0                                            NaN              0
1  What you will get from usThrough being part of...              0
2                                            NaN              0

   has_company_logo  has_questions employment_type required_experience  \
0                 1             0           Other          Internship
1                 1             0       Full-time      Not Applicable
2                 1             0             NaN                 NaN

  required_education                  industry          function  fraudulent
0                NaN                       NaN         Marketing           0
1                NaN  Marketing and Advertising  Customer Service           0
2                NaN                       NaN               NaN           0
```

LAST 3 ROWS :

```
--- Last 3 Rows ---
       job_id                                                title  \
17877   17878  Project Cost Control Staff Engineer - Cost Con...
17878   17879                                  Graphic Designer
17879   17880                         Web Application Developers

               location   department salary_range  \
17877    US, TX, Houston          NaN          NaN
17878      NG, LA, Lagos          NaN          NaN
17879  NZ, N, Wellington  Engineering          NaN

                                       company_profile  \
17877  We Provide Full Time Permanent Positions for m...
17878                                              NaN
17879  Vend is looking for some awesome new talent to...

                                           description  \
17877  Experienced Project Cost Control Staff Enginee...
17878  Nemsia Studios is looking for an experienced v...
17879  Who are we?Vend is an award winning web based ...

                                          requirements  \
17877  At least 12 years professional experience.Abil...
17878  1. Must be fluent in the latest versions of Co...
17879  We want to hear from you if:You have an in-dep...

                                            benefits  telecommuting  \
17877                                            NaN              0
17878  Competitive salary (compensation will be based...              0
17879                                            NaN              0

      has_company_logo  has_questions employment_type required_experience  \
17877                0              0       Full-time                 NaN
17878                0              1        Contract      Not Applicable
17879                1              1       Full-time    Mid-Senior level

     required_education           industry     function  fraudulent
17877                NaN                NaN          NaN           0
17878      Professional     Graphic Design       Design           0
17879                NaN  Computer Software  Engineering           0
```

4.2 CLASSIFICATION AND SUMMARY REPORT :

```
==========================================
MODEL PERFORMANCE
==========================================

Classification Report:
              precision    recall  f1-score   support

    Not Scam       0.97      1.00      0.98      3402
        Scam       1.00      0.29      0.46       173

    accuracy                           0.97      3575
   macro avg       0.98      0.65      0.72      3575
weighted avg       0.97      0.97      0.96      3575

Confusion Matrix:
                Predicted
               0      1
Actual  0    | 3402   0
        1    | 122    51
 ====================================
 SUMMARY REPORT
 ====================================
 Dataset after cleaning: 17871 rows, 2 columns
 Columns used for training: description, fraudulent

 Class distribution:
 fraudulent
 0    17006
 1      865
 Name: count, dtype: int64

 Model: Logistic Regression
 Vectorization: TF-IDF (stop_words='english', max_df=0.7)
 Train/Test split: 80% train, 20% test

 Performance:
 Accuracy: 96.59%
 True Negatives: 3402, False Positives: 0
 False Negatives: 122, True Positives: 51

 Interpretation: The model performs well in detecting fake job postings.
```

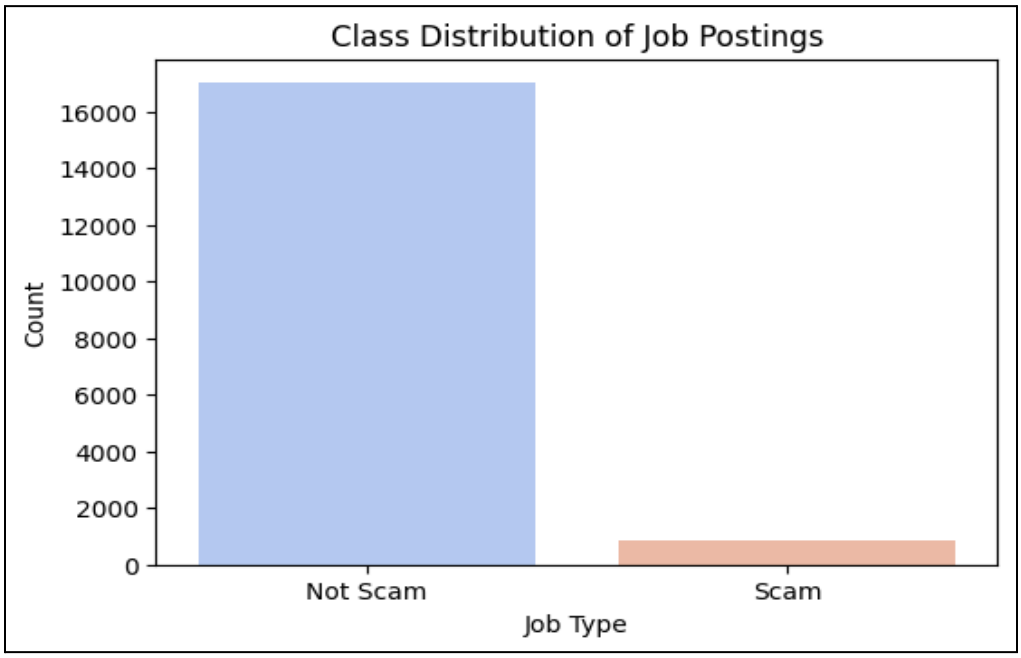
4.3 EDA VISUALIZATION REPORT :

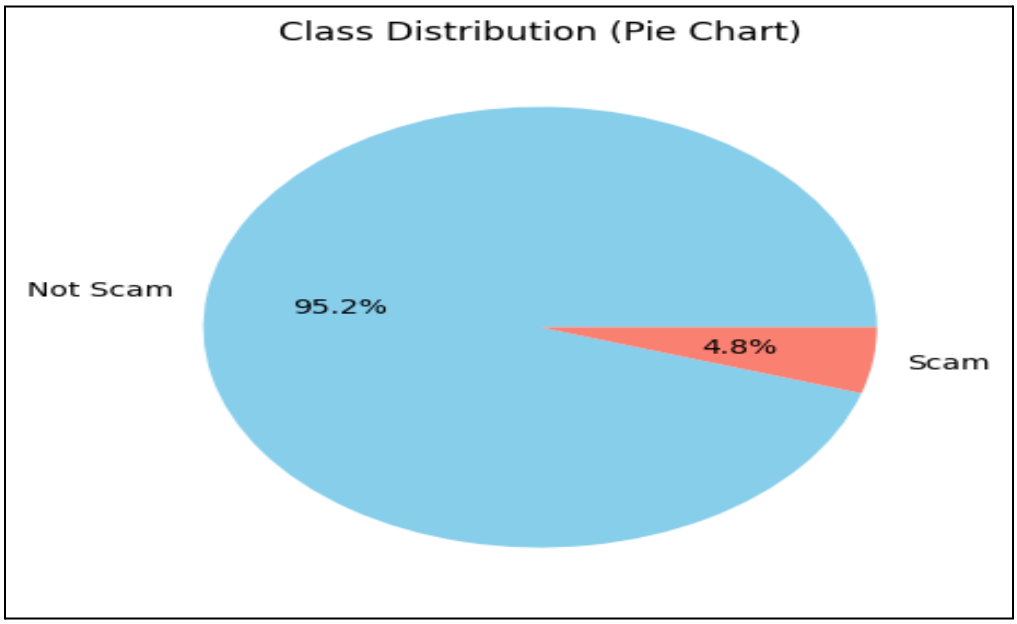
1. Class Distribution

Objective: To examine the proportion of scam and non-scam postings in the dataset, providing an overview of class imbalance.

Plots:

- Count plot (bar chart)
- Pie chart



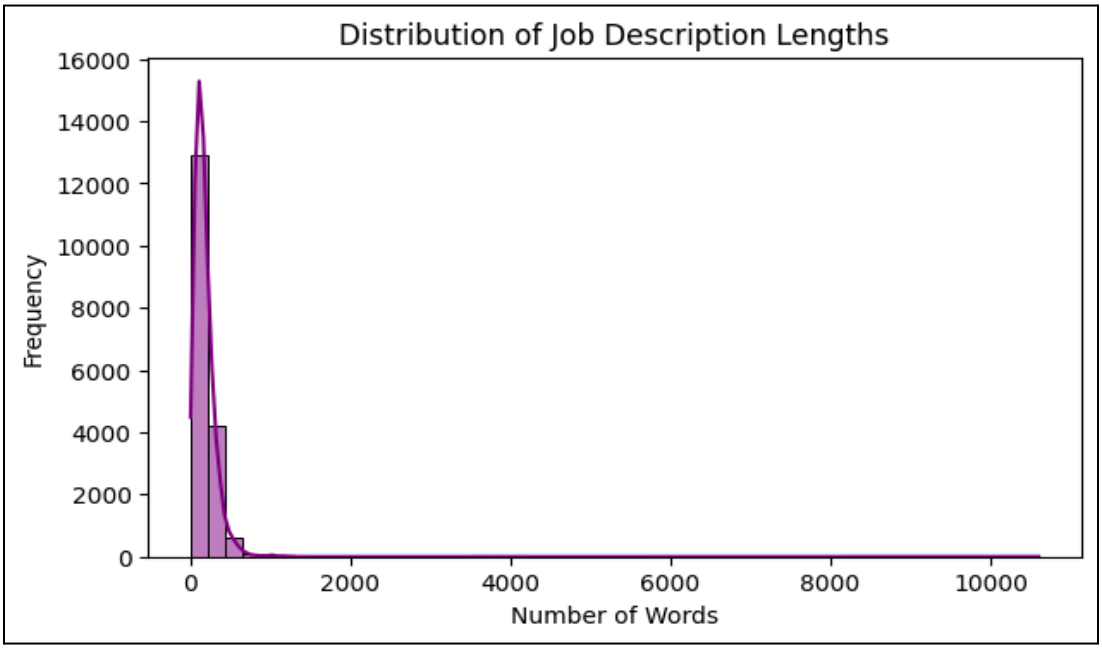


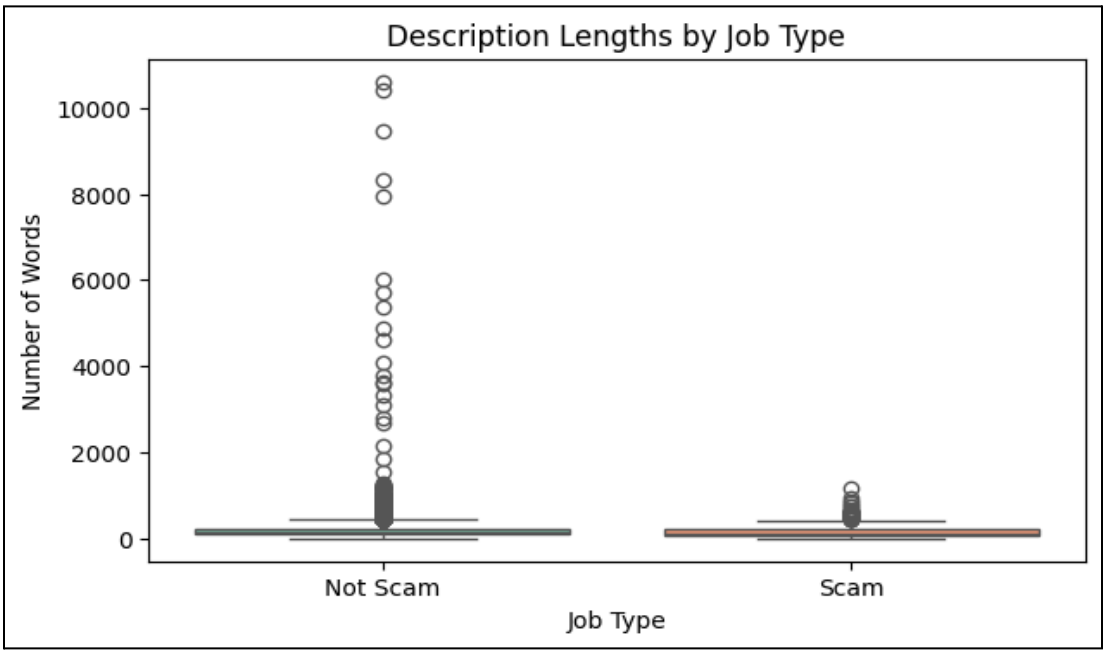
2. Length-Based Analysis

Objective: To analyze the distribution of job description lengths and identify differences in average length between scam and non-scam postings.

Plots:

- Histogram of lengths
- Box plot comparing scam vs non-scam lengths

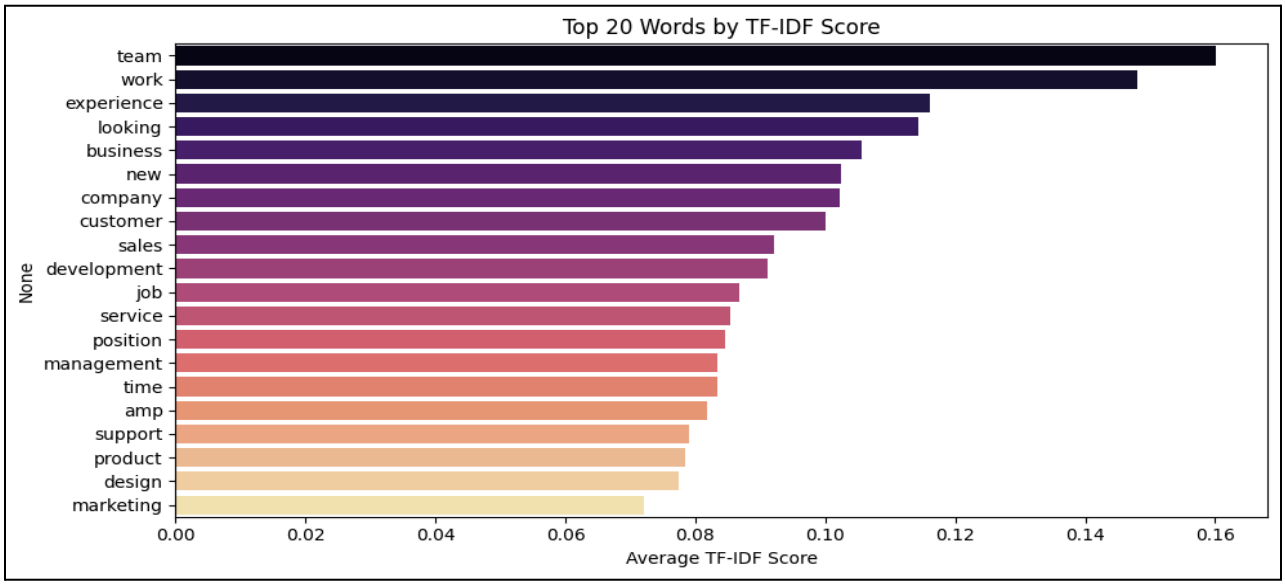


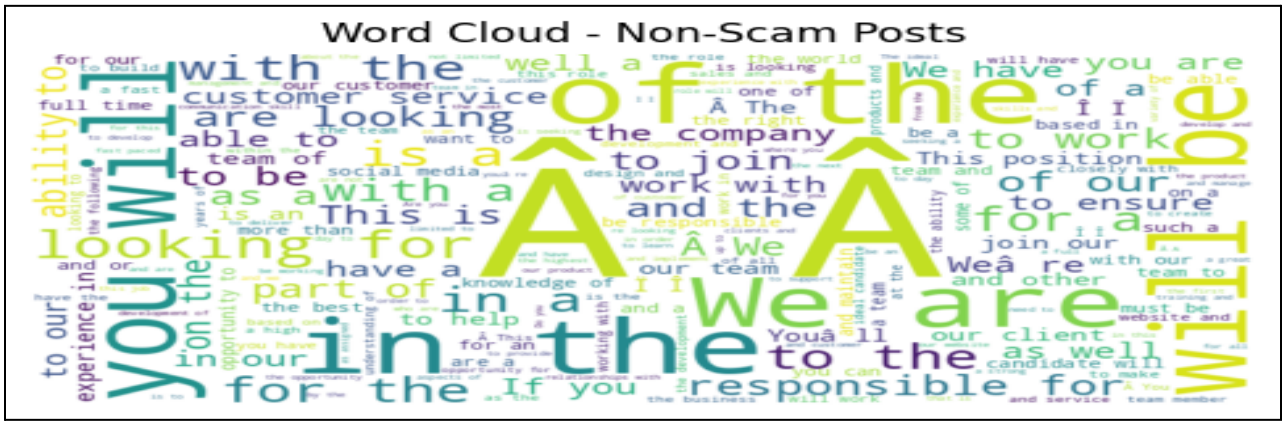


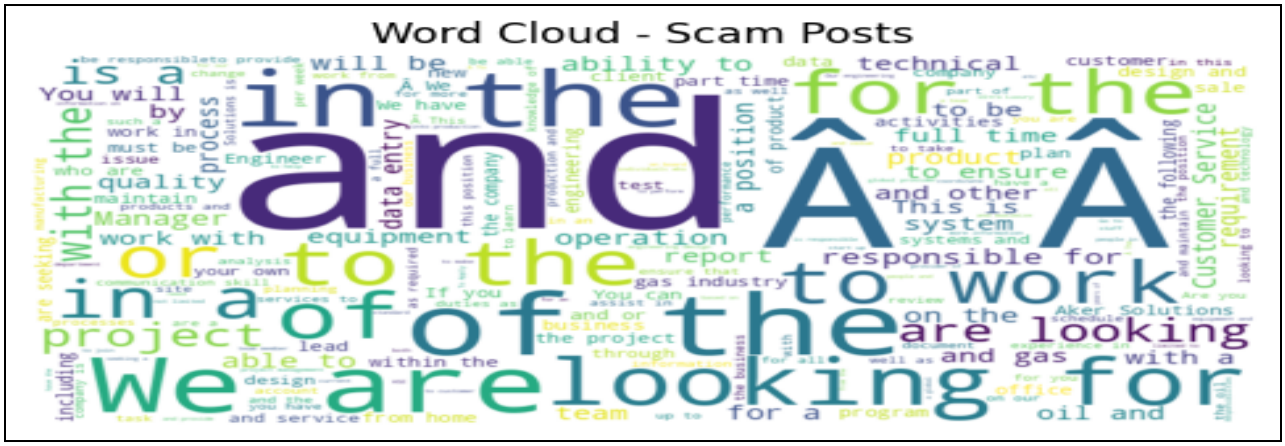
3. Word Frequency Analysis

Objective: To identify the most frequently occurring words in scam and non-scam postings, highlighting potential keyword patterns.
Plots:

- Horizontal bar charts for top words
- Word clouds







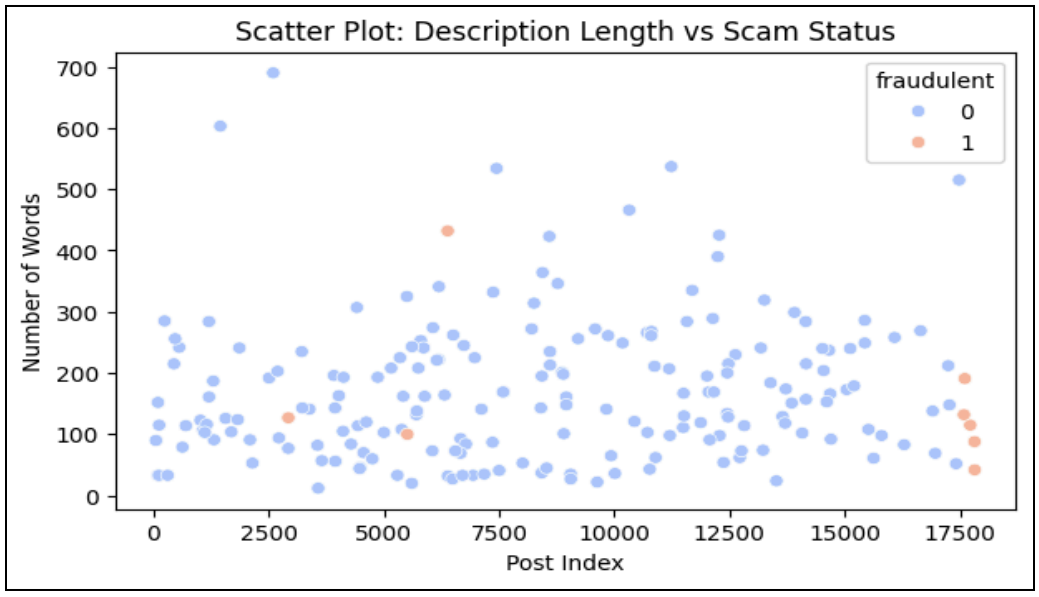
4.Length–Scam Relationship

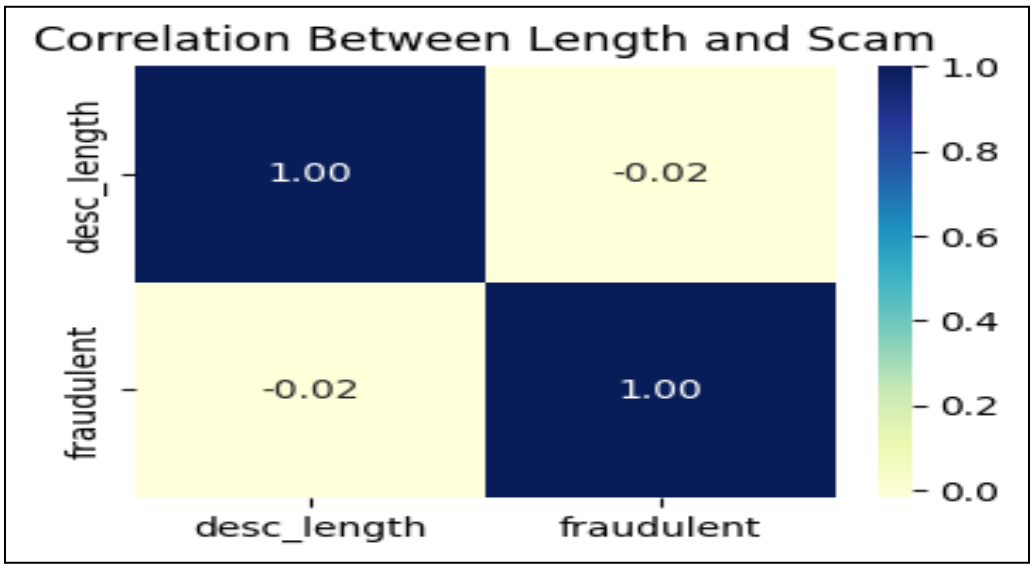
Objective: To explore how job description length correlates with the likelihood of a posting being classified as a scam.

Plots:

- Scatter plot (sampled for clarity)

- Correlation heatmap

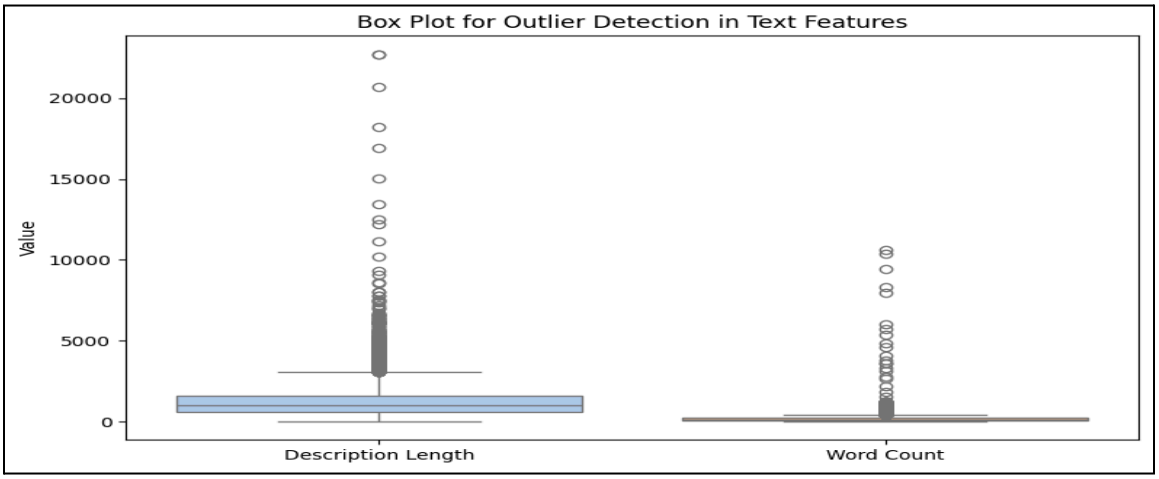


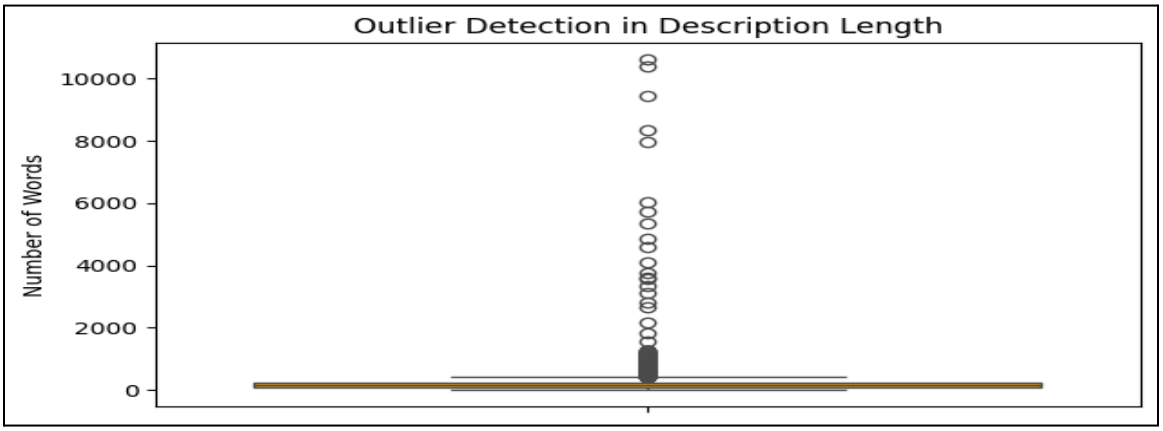


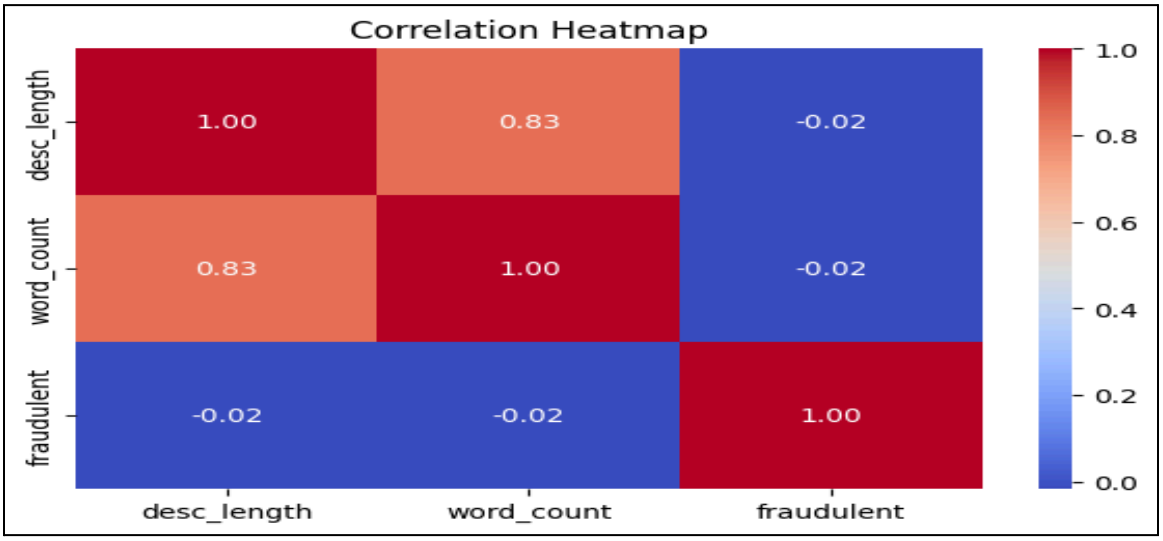
5. Outlier Detection

Objective: To detect and visualize unusual data points that significantly deviate from normal patterns, which may indicate data quality issues or atypical scam behavior.
Plots:

- Box plot highlighting outliers
- Correlation heat map







CONCLUSION

My internship experience at Eagle Hi-Tech Softclou Pvt Ltd has been immensely valuable and rewarding. As a college student, the opportunity to work on a real-world machine learning project using Google Colab provided me with hands-on experience and practical skills that complemented my academic learning.

During my time at Eagle Hi-Tech Softcloud Pvt Ltd, I actively worked on a project focused on **Detection of Deceptive Job Postings Using Machine Learning**. The process involved data preprocessing, exploratory data analysis, feature extraction using TF-IDF, and building a predictive model with Logistic Regression to classify postings as genuine or fake.

This experience enhanced my understanding of natural language processing techniques, model evaluation, and the challenges of working with unstructured text data in real-world scenarios.

This internship has reinforced my passion for data science and machine learning while deepening my appreciation for the role of data-driven solutions in enhancing the safety and trustworthiness of online job platforms. The real-world exposure has strengthened my technical expertise in NLP, model building, and evaluation, while boosting my problem-solving abilities and confidence in applying analytical methods to practical challenges.

I am grateful to Eagle Hi-Tech Softclou Pvt Ltd for providing me with this exceptional opportunity and to my college for facilitating such an enriching internship experience. I look forward to applying the knowledge and expertise gained during this internship to make meaningful contributions to future projects and organizations.